Hailey Hudson

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Education

New Mexico State University Expected December 2026
Bachelor of Science in Computer Science, Minor in Government GPA: 3.5
Relevant Coursework: Human-Centered Computing, Data Structures & Algorithms, Database Management Systems (SQL), Operating Systems, Parallel Programming, Object-Oriented Programming

Research Experience

Undergraduate Research Assistant Jan. 2026 – May 2026
New Mexico State University Las Cruces, NM

- Designed 3 guided exercises with learning objectives and evaluation rubrics for a Docker-containerized Honeyd honeypot lab on cyber defense and deception to make cyber deception education more accessible
- Presented a research poster on Lowering the Barrier to Cyber Deception Education with Containerized Honeypot Labs at the CRA UR2PhD Development Workshop in New Orleans, LA

Projects

StoryWeave: Adaptive Story Generator for Early Readers (Capstone) | *React, TypeScript, Python* Sept. 2026 – Present

- Analyzed 2,474 App Store and Google Play reviews across 5 reading apps using a Python scraper (requests, google-play-scraper, pandas, TextBlob) with retry logic for unreliable endpoints, sentiment-scoring each review
- Identified age-inappropriate and repetitive content as the top complaints and translated them into design requirements for the story generator
- Built the React/TypeScript read-aloud screen and caregiver dashboard on a 3-person Scrum team

WaterOptix: Drought Adaptation Manager | *React, TypeScript, Python, FastAPI, REST* April 2026

- Interviewed 3 local farmers and an NMSU agricultural professor to shape the tool around seasonal planting decisions
- Built the React/TypeScript frontend for a full-stack drought decision tool in a 36-hour hackathon (2nd place, team of 3), connecting 2 FastAPI REST endpoints to display 0–60% shortage forecasts for 8 crops

Movie Tinder: Movie Recommendation System | *JavaScript, React, TensorFlow, MongoDB* Jan. 2026 – May 2026

- Owned the TMDB API integration and MongoDB data layer for a full-stack movie discovery app with a swipe interface, on a team of 4
- Designed the MongoDB schema and trained a TensorFlow model on user swipe data to personalize recommendations

Voice Reminders | *React Native, Expo, TypeScript* Sept. 2025 – Dec. 2025

- Built audio recording and playback in Expo and redesigned the playback layout after usability testing the iPhone prototype with 10 peers

Teaching & Leadership

Tech Fellow August 2026 – Present
CodePath Las Cruces, NM

- Lead office hours for 50 intro CS students, debugging their code and explaining data structures and control flow

Student Staff May 2025 – Present
Young Women in Computing Las Cruces, NM

- Taught programming fundamentals to 45 middle and high school students with no prior experience across 10 sessions

Technical Skills

Languages: Java, Python, SQL, TypeScript, JavaScript, C++, C
Web & APIs: React, React Native, Expo, Node.js, FastAPI, REST APIs, HTML/CSS
Databases: PostgreSQL, MySQL (relational); MongoDB (NoSQL)
Data & ML: TensorFlow, scikit-learn, pandas, NumPy, TextBlob
Research Methods: User interviews, usability testing, sentiment analysis, app-review mining
Tools & Practices: Agile/Scrum, Git/GitHub (pull requests), JUnit, Docker, Linux, SSH, pthreads; AI-assisted development with code review and debugging

Honors

Great Minds in STEM Data Analytics Challenge, 3rd of 20 teams Oct. 2025

- Tuned scikit-learn regression and classification models in Python to reach an R-squared of 0.984 on a real-world dataset with a team of 3

CAHSI Student Scholar Award 2026

- Recognized for efforts in community outreach, campus involvement, and academic performance